

CSEB1122T: Computer Programming

Time Allowed: 1 Hr

Max. Marks: 15

Note: Attempt all questions in section-A and attempt *any-two* from section-B.

SECTION-A

- Q-1 Perform conversion from the given number into corresponding base number systems:
- (73456.321)₈ to (?)₁₆ (?)₁₀ and (?)₂
 - (13CCA.B55)₁₆ to (?)₁₀ (?)₂ and (?)₈
 - (62123)₆ to (?)₁₀ and (?)₂
- (b) Given binary A=11001.011 and B=1101.11. Perform A×B using computer method.
- (c) Subtract (1010)₂ from (1011)₂ using additive method with 2's complement.
- (d) Define the terms of '*identifiers*' and '*keywords*'. Give examples of each how these are used in 'C' language.

SECTION-B

- Q2 i) To iterate any one statement/expression in your 'C' program, how many 'iterative methods' are available to use. Give one example of each.
- ii) Discuss with examples '*decision-making-structures*' used in 'C'.
- Q3 i) Give examples of '*auto*' and '*static*' variables of primitive data types and discuss how these are different with each other.
- ii) Using '*conditional operator*', compare three integer numbers X, Y and Z.
- Q4. From a series of integers from 1 to 25, a user wants to compute product of each number 'multiplied by 3 and fully divisible by 4', otherwise compute sum when number is fully divisible by 3. Write a procedure in 'C' for the same.
- Q5. Explain the use of '*Assembler*', '*Compiler*' and '*interpreter*' in programming. How '*low-level*' programming is different than '*high-level*' languages. Justify with suitable examples.

BTech – 1stSem
Subject: Computer Programming

MST-1
Subject Code: CPE-101 Max Marks: 15

Time: 1 hour

Department of Computer Engineering, Punjabi University, Patiala

Section –A

1. Define Following Terms : (5 Marks)

(a). Compiler (b). Escape Sequences (c). Keyword (d). Variable (e). Bitwise & operator

Section –B (Do any Two Questions)

2. Explain various types of loops with syntax and example. (5 Marks)

3. Perform: (5 Marks)

a) $1110.11_{(2)} \rightarrow __(10), __(8), __(16)$

b) $1234_{(10)} \rightarrow __(2), __(8), __(16)$

c) $A0B1_{(16)} \rightarrow __(10), __(2), __(8)$

d) $110111 + 11011 + 101010$

4. a) Write a Program in “C” language to input a number print it is even or odd. (2 Marks)

b) Write a Program in “C” language to input a number print it is prime or not (3 Marks)

8/4/137
50

12/01/078

Brem
Keimar

6481
10

Time Allowed: 60 Minutes

Max. Marks: 15

Question 1 is Compulsory. Attempt *any two* from section-B. Marks of each question give against each.

SECTION-A

- Q.1 i) Perform following number conversions: $(\frac{1}{2} + \frac{1}{2}) = 1$
- | | | |
|---|-------------------------------------|-----|
| (a) $(58FDC)_{16} \rightarrow (?)_8 \rightarrow (?)_{10}$ | (b) $(9786)_{10} \rightarrow (?)_5$ | (5) |
| vi) Multiply $(1101)_2$ to $(1011)_2$ by computer method. | 1 | |
| vii) Why do we need header file in a C program | 1 | |
| viii) Define <i>multiprogramming</i> vs Multitasking operating system | 1 | |
| ix) What is the difference between & and && operator | 1 | |

SECTION-B

- Q.2 In a 'C' program different variable declared as **int**, **short int**, **long int**, **signed int**, **unsigned int**, **long**, **double**, **long double**, **char**, **short char**. Give their memory requirement for each type of one variable in bits/bytes and values ranges. (5)
- Q.3 Write a program in 'C' to perform following operations on integers between 1-50: (5)
- | | |
|--|--|
| iii) Sum of all even elements divisible only by 3. | |
| iv) Product of all odd by adding 4 in each element. | |
- Q.4 Discuss repetitive constructs/loops used in 'C' programming. How you decide to prefer one over others. Justify your answer with appropriate example. (5)